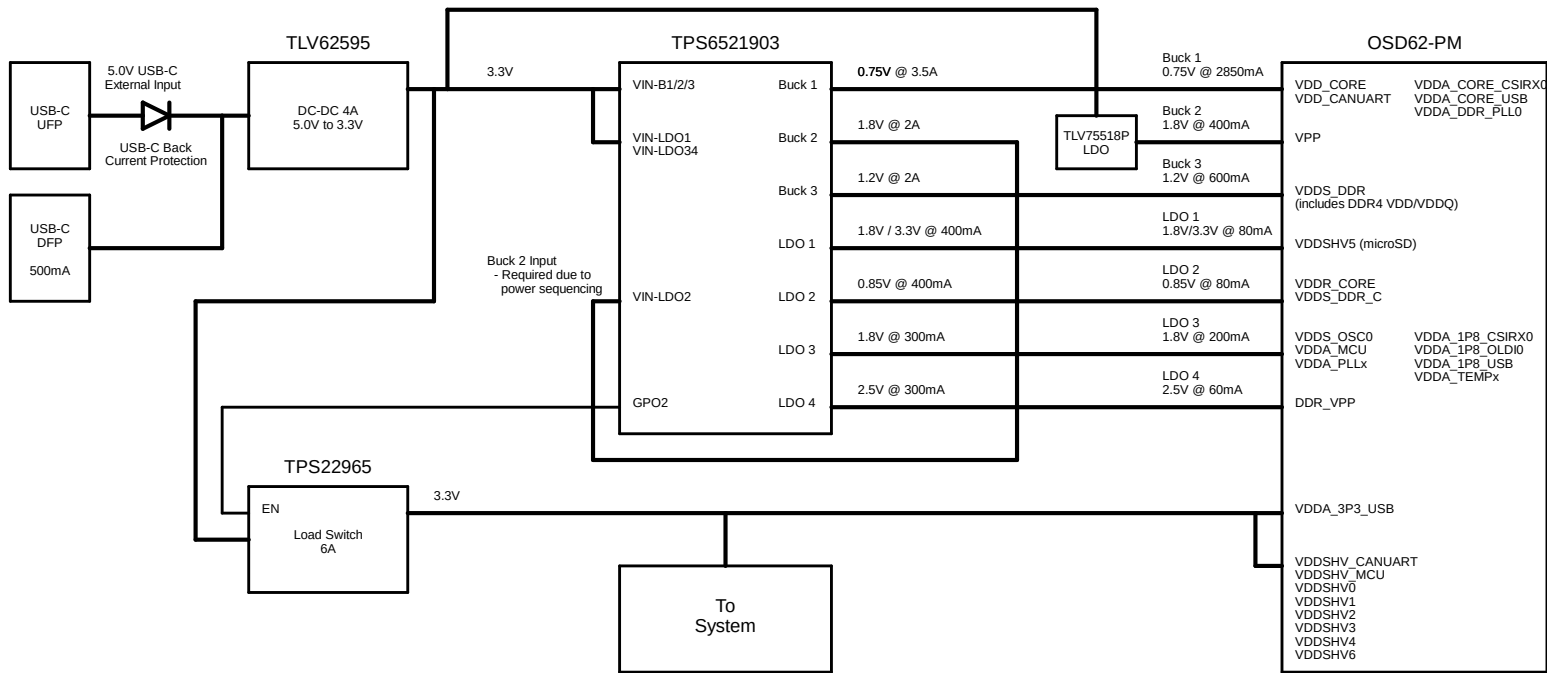


|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |         |   |   |                                                                                                                                                                                                                                                                                                                  |                |                                             |                         |                                                                                       |                           |                                       |                           |                                            |                     |                          |                 |                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |  |  |  |                             |                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
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| A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <table><tr><td>Table of Contents</td></tr><tr><td>Sheet 1: Title</td></tr><tr><td>Sheet 2: Notes / Block Diagram / Mechanical</td></tr><tr><td>Sheet 3: OSD62-PM Power</td></tr><tr><td>Sheet 4: OSD62-PM GND / IO Power</td></tr><tr><td>Sheet 5: OSD62-PM Signals</td></tr><tr><td>Sheet 6: USB-C / CSI / Power In / Osc</td></tr><tr><td>Sheet 7: Power Management</td></tr><tr><td>Sheet 8: JTAG / Boot Mode / microSD / etc.</td></tr><tr><td>Sheet 9: IO Headers</td></tr><tr><td>Sheet 10: LVDS Connector</td></tr><tr><td>Sheet 11: Blank</td></tr><tr><td>Sheet 12: Design Notes</td></tr></table> |         |   |   | Table of Contents                                                                                                                                                                                                                                                                                                | Sheet 1: Title | Sheet 2: Notes / Block Diagram / Mechanical | Sheet 3: OSD62-PM Power | Sheet 4: OSD62-PM GND / IO Power                                                      | Sheet 5: OSD62-PM Signals | Sheet 6: USB-C / CSI / Power In / Osc | Sheet 7: Power Management | Sheet 8: JTAG / Boot Mode / microSD / etc. | Sheet 9: IO Headers | Sheet 10: LVDS Connector | Sheet 11: Blank | Sheet 12: Design Notes | <table><tr><td>Octavo Systems OSD62-PM-BRK</td></tr><tr><td>(c) Copyright, Octavo Systems LLC, 2024-2025. All Rights Reserved.</td></tr><tr><td>THERE IS NO WARRANTY FOR THIS DESIGN, TO THE EXTENT PERMITTED BY APPLICABLE LAW. EXCEPT WHEN OTHERWISE STATED IN WRITING THE COPYRIGHT HOLDERS AND/OR OTHER PARTIES PROVIDE THE DESIGN *AS IS* WITHOUT WARRANTY OF ANY KIND, EITHER EXPRESSED OR IMPLIED, INCLUDING, BUT NOT LIMITED TO, THE IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS A FOR A PARTICULAR PURPOSE. THE ENTIRE RISK AS TO THE QUALITY AND PERFORMANCE OF THE DESIGN IS WITH YOU. SHOULD THE DESIGN PROVE DEFECTIVE, YOU ASSUME THE COST OF ALL NECESSARY SERVICING, REPAIR OR CORRECTION.</td></tr></table> |  |  |  | Octavo Systems OSD62-PM-BRK | (c) Copyright, Octavo Systems LLC, 2024-2025. All Rights Reserved. | THERE IS NO WARRANTY FOR THIS DESIGN, TO THE EXTENT PERMITTED BY APPLICABLE LAW. EXCEPT WHEN OTHERWISE STATED IN WRITING THE COPYRIGHT HOLDERS AND/OR OTHER PARTIES PROVIDE THE DESIGN *AS IS* WITHOUT WARRANTY OF ANY KIND, EITHER EXPRESSED OR IMPLIED, INCLUDING, BUT NOT LIMITED TO, THE IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS A FOR A PARTICULAR PURPOSE. THE ENTIRE RISK AS TO THE QUALITY AND PERFORMANCE OF THE DESIGN IS WITH YOU. SHOULD THE DESIGN PROVE DEFECTIVE, YOU ASSUME THE COST OF ALL NECESSARY SERVICING, REPAIR OR CORRECTION. |
| Table of Contents                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |         |   |   |                                                                                                                                                                                                                                                                                                                  |                |                                             |                         |                                                                                       |                           |                                       |                           |                                            |                     |                          |                 |                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |  |  |  |                             |                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Sheet 1: Title                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |         |   |   |                                                                                                                                                                                                                                                                                                                  |                |                                             |                         |                                                                                       |                           |                                       |                           |                                            |                     |                          |                 |                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |  |  |  |                             |                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
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| Sheet 5: OSD62-PM Signals                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |         |   |   |                                                                                                                                                                                                                                                                                                                  |                |                                             |                         |                                                                                       |                           |                                       |                           |                                            |                     |                          |                 |                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |  |  |  |                             |                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
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| Sheet 7: Power Management                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |         |   |   |                                                                                                                                                                                                                                                                                                                  |                |                                             |                         |                                                                                       |                           |                                       |                           |                                            |                     |                          |                 |                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |  |  |  |                             |                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
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| Sheet 9: IO Headers                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |         |   |   |                                                                                                                                                                                                                                                                                                                  |                |                                             |                         |                                                                                       |                           |                                       |                           |                                            |                     |                          |                 |                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |  |  |  |                             |                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Sheet 10: LVDS Connector                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |         |   |   |                                                                                                                                                                                                                                                                                                                  |                |                                             |                         |                                                                                       |                           |                                       |                           |                                            |                     |                          |                 |                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |  |  |  |                             |                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Sheet 11: Blank                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |         |   |   |                                                                                                                                                                                                                                                                                                                  |                |                                             |                         |                                                                                       |                           |                                       |                           |                                            |                     |                          |                 |                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |  |  |  |                             |                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Sheet 12: Design Notes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |         |   |   |                                                                                                                                                                                                                                                                                                                  |                |                                             |                         |                                                                                       |                           |                                       |                           |                                            |                     |                          |                 |                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |  |  |  |                             |                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Octavo Systems OSD62-PM-BRK                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |         |   |   |                                                                                                                                                                                                                                                                                                                  |                |                                             |                         |                                                                                       |                           |                                       |                           |                                            |                     |                          |                 |                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |  |  |  |                             |                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
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| THERE IS NO WARRANTY FOR THIS DESIGN, TO THE EXTENT PERMITTED BY APPLICABLE LAW. EXCEPT WHEN OTHERWISE STATED IN WRITING THE COPYRIGHT HOLDERS AND/OR OTHER PARTIES PROVIDE THE DESIGN *AS IS* WITHOUT WARRANTY OF ANY KIND, EITHER EXPRESSED OR IMPLIED, INCLUDING, BUT NOT LIMITED TO, THE IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS A FOR A PARTICULAR PURPOSE. THE ENTIRE RISK AS TO THE QUALITY AND PERFORMANCE OF THE DESIGN IS WITH YOU. SHOULD THE DESIGN PROVE DEFECTIVE, YOU ASSUME THE COST OF ALL NECESSARY SERVICING, REPAIR OR CORRECTION. |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |         |   |   |                                                                                                                                                                                                                                                                                                                  |                |                                             |                         |                                                                                       |                           |                                       |                           |                                            |                     |                          |                 |                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |  |  |  |                             |                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
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| C                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |         |   |   | OSD62-PM BRK                                                                                                                                                                                                                                                                                                     |                |                                             |                         |                                                                                       |                           |                                       |                           |                                            |                     |                          |                 |                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |  |  |  |                             |                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| D                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |         |   |   | Version 1D                                                                                                                                                                                                                                                                                                       |                |                                             |                         |                                                                                       |                           |                                       |                           |                                            |                     |                          |                 |                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |  |  |  |                             |                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| E                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |         |   |   | <table><tr><td rowspan="4"></td><td colspan="2">Octavo Systems LLC.</td></tr><tr><td colspan="2">OSD62-PM-BRK</td></tr><tr><td colspan="2">7/15/2025 5:40 PM</td></tr><tr><td>Sheet: 1/12</td><td>Rev: 1D</td></tr></table> |                |                                             |                         |  | Octavo Systems LLC.       |                                       | OSD62-PM-BRK              |                                            | 7/15/2025 5:40 PM   |                          | Sheet: 1/12     | Rev: 1D                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |  |  |  |                             |                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Octavo Systems LLC.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |         |   |   |                                                                                                                                                                                                                                                                                                                  |                |                                             |                         |                                                                                       |                           |                                       |                           |                                            |                     |                          |                 |                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |  |  |  |                             |                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | OSD62-PM-BRK                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |         |   |   |                                                                                                                                                                                                                                                                                                                  |                |                                             |                         |                                                                                       |                           |                                       |                           |                                            |                     |                          |                 |                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |  |  |  |                             |                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
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|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Sheet: 1/12                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Rev: 1D |   |   |                                                                                                                                                                                                                                                                                                                  |                |                                             |                         |                                                                                       |                           |                                       |                           |                                            |                     |                          |                 |                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |  |  |  |                             |                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
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# Power Block Diagram



## USB-C Power Discussion

The USB-C UFP uses 5.1k pull down resistors on the CC pins to request up to 3A from the DFP. However, this board does not check the voltage of the CC pins to understand the current provided by the DFP port. The USB-C DFP port on this board uses 56k ohm pull up resistors to advertise 500mA of current available. It is the responsibility of the user to guarantee that there is enough current to support both the components on the board as well as any downstream USB devices.

## I2C Table

| I2C0    |            |
|---------|------------|
| ADDRESS | DEVICE     |
| 0x13    | TPS65219   |
| 0x50    | EEPROM     |
| 0x51    | ADC121C027 |
| 0x52    | ADC121C027 |

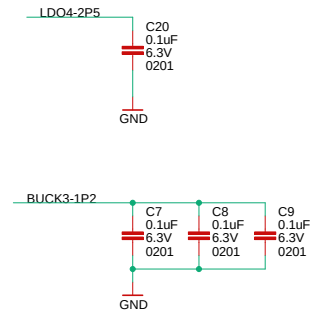
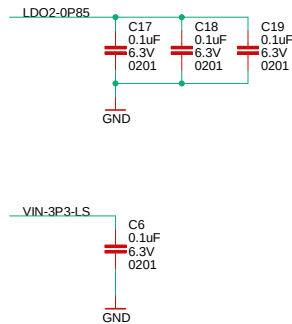
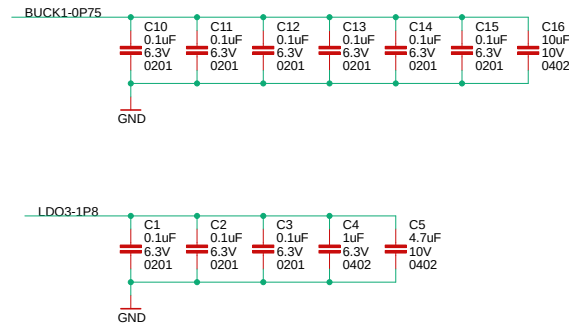
## Mechanical & Misc

### Fiducials



|                     |         |
|---------------------|---------|
| Octavo Systems LLC. |         |
| OSD62-PM-BRK        |         |
| 7/15/2025 5:40 PM   |         |
| Sheet: 2/12         | Rev: 1D |

## Power Connections for the OSD62-PM

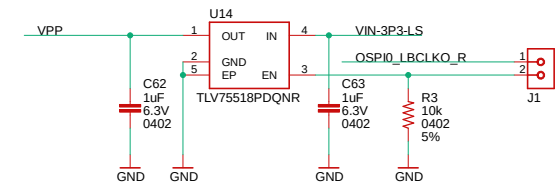
Internal PM / OSD62 BRK Board

- <https://www.ti.com/lit/an/sprac76f/sprac76f.pdf> (see page 29)
- <https://www.ti.com/lit/zip/spar001> (See Sheet 14)

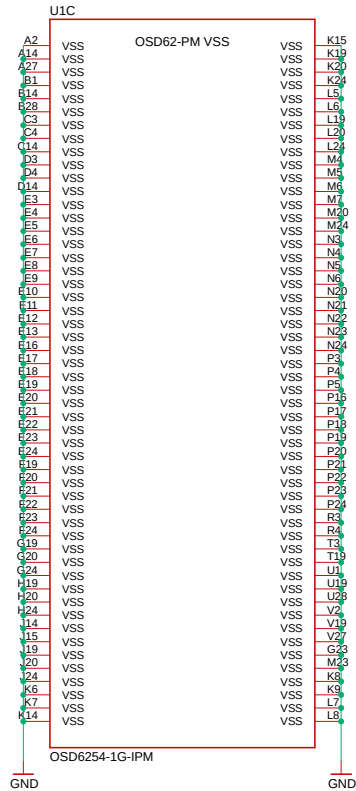
- <https://www.ti.com/lit/an/sprada6/sprada6.pdf>

To program internal OTP memory (security keys, etc.), VPP must be supplied by an externally controllable power supply.

Jumper J1 must be installed to program the OTP memory.



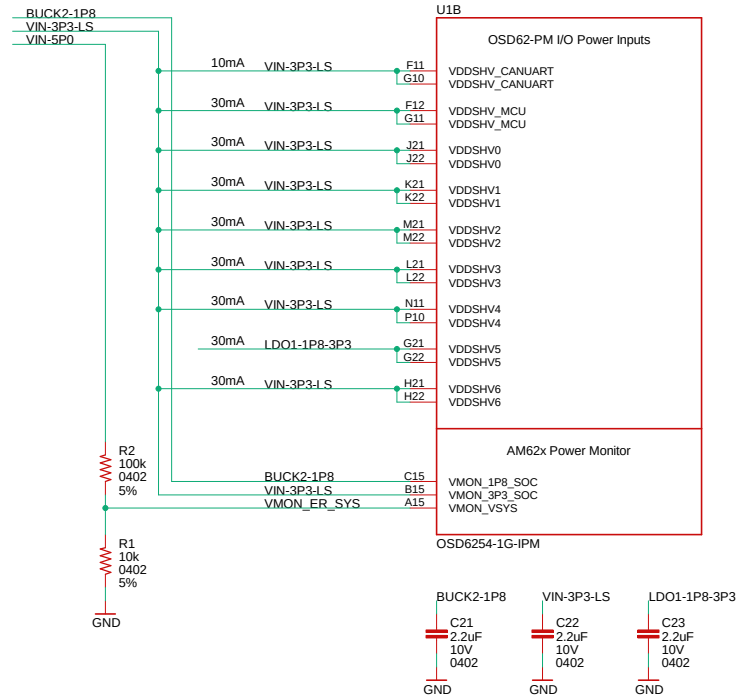
## OSD62-PM Ground



## OSD62-PM I/O Power

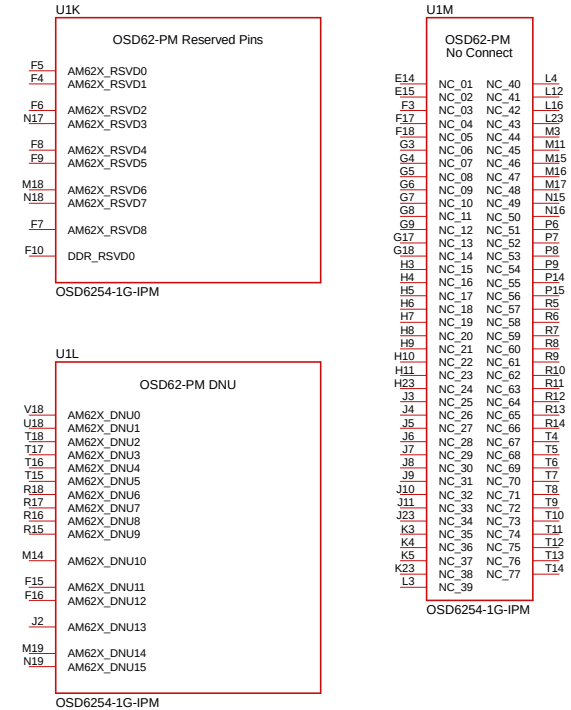
All IO to headers on OSD62-PM are 3.3V.

Other designs will need to set VDDSHV banks to meet their IO voltage needs.

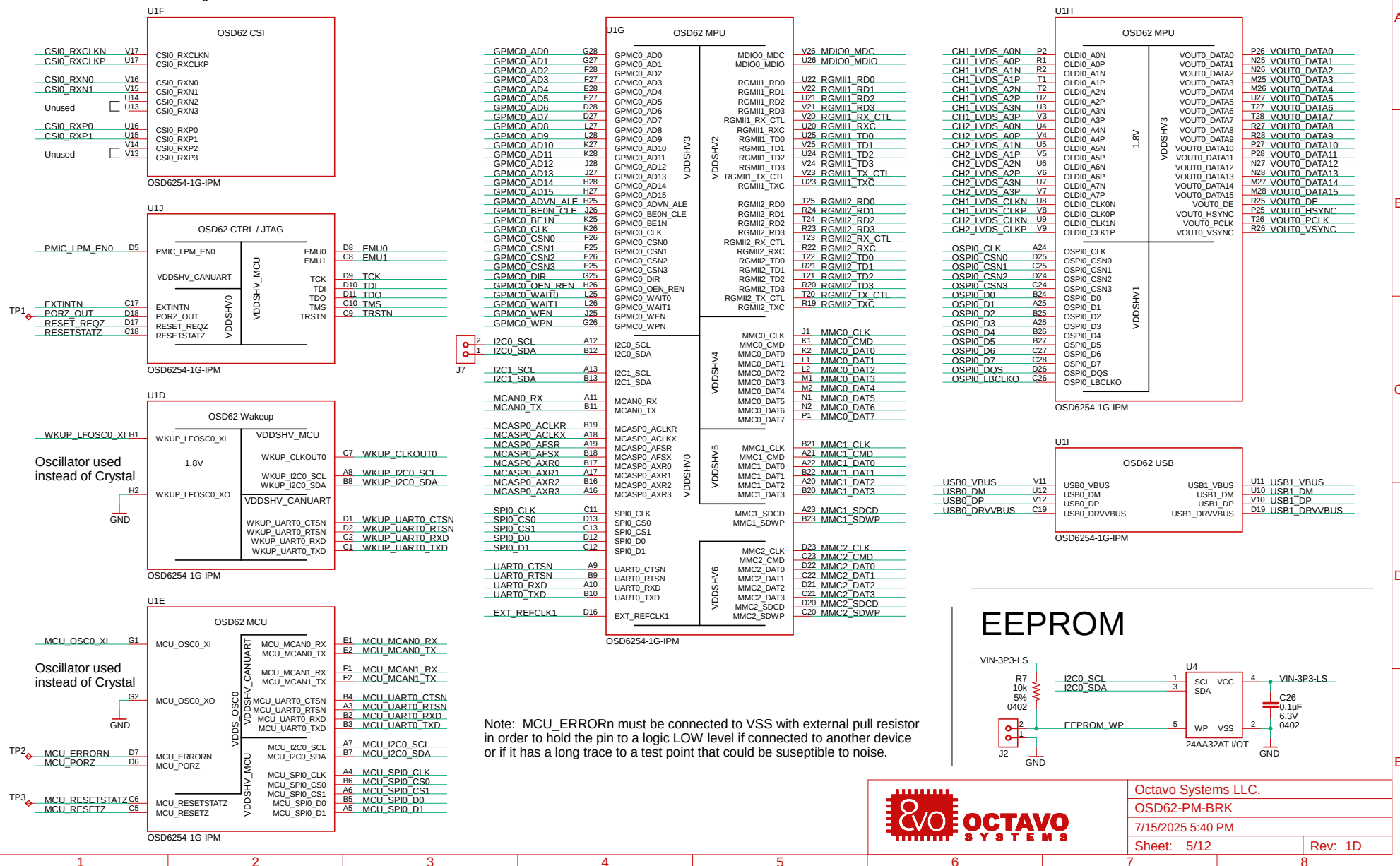


## OSD62-PM Reserved / DNU

These pins are reserved and should not be connected or routed thru.



Test Points for unused control signals.



Octavo Systems LLC.

OSD62-PM-BRK

7/15/2025 5:40 PM

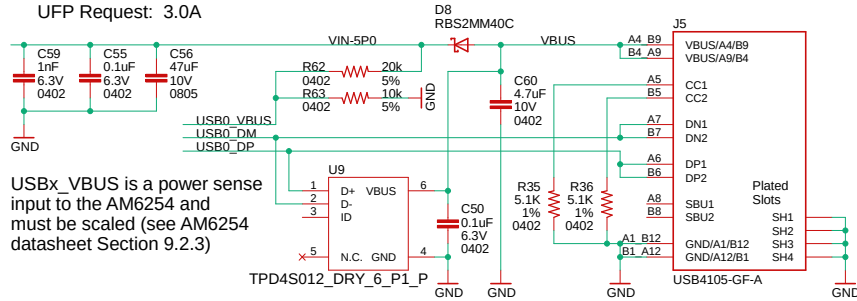
Sheet: 5/12

Rev: 1D

## USB-C Connector

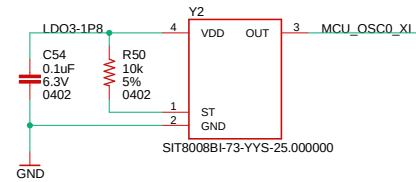
Only supports USB2.0; Cannot supply power  
UFP Request: 3.0A

USB-C Back Current Protection  
Allows USB connection when  
board is externally powered.  
(Max Vf = 400mV @ 2A)

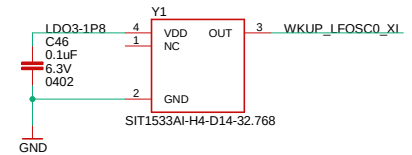


## Oscillators

OSD62x 25 MHz Oscillator  
1.8V only



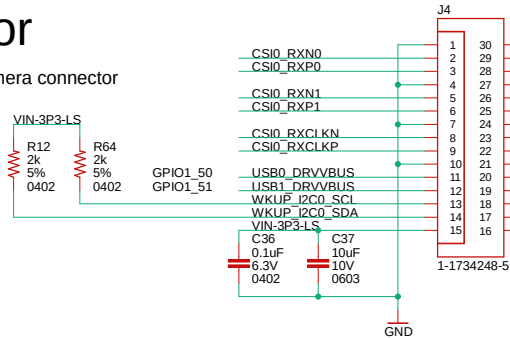
OSD62x 32 kHz Oscillator  
1.8V only



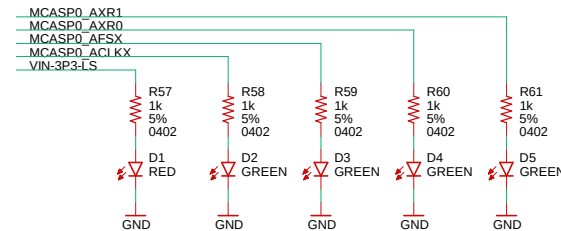
## CSI Connector

Compatible with Raspberry Pi camera connector

CSI length requirements:  
- Inter pair: 0.15mm  
- Intra pair: 1.5mm

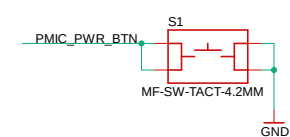


## LEDs



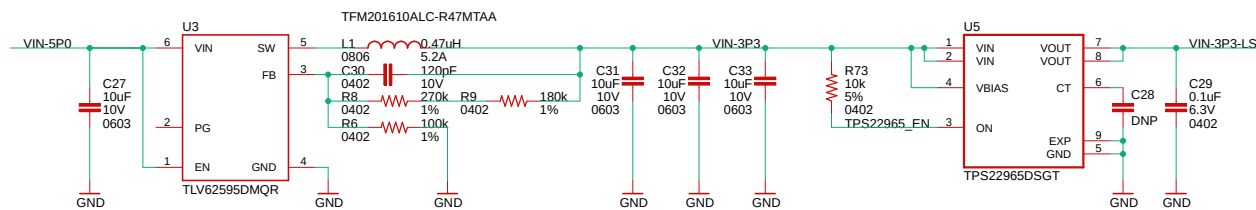
USR2 USR3 USR1 USR0

## Power Button



## 3.3V DC-DC Converter

Main 3.3V Input

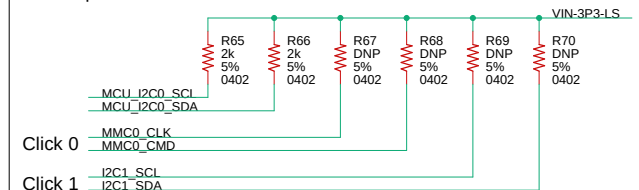


4A DC-DC converter

6A Load Switch

## I2C Pull Ups

Pull up resistors are required for the MCU and WKUP I2C busses regardless of use.  
No pull ups required for the Click I2C Interfaces, per spec.  
Footprints on back of board added for convenience.



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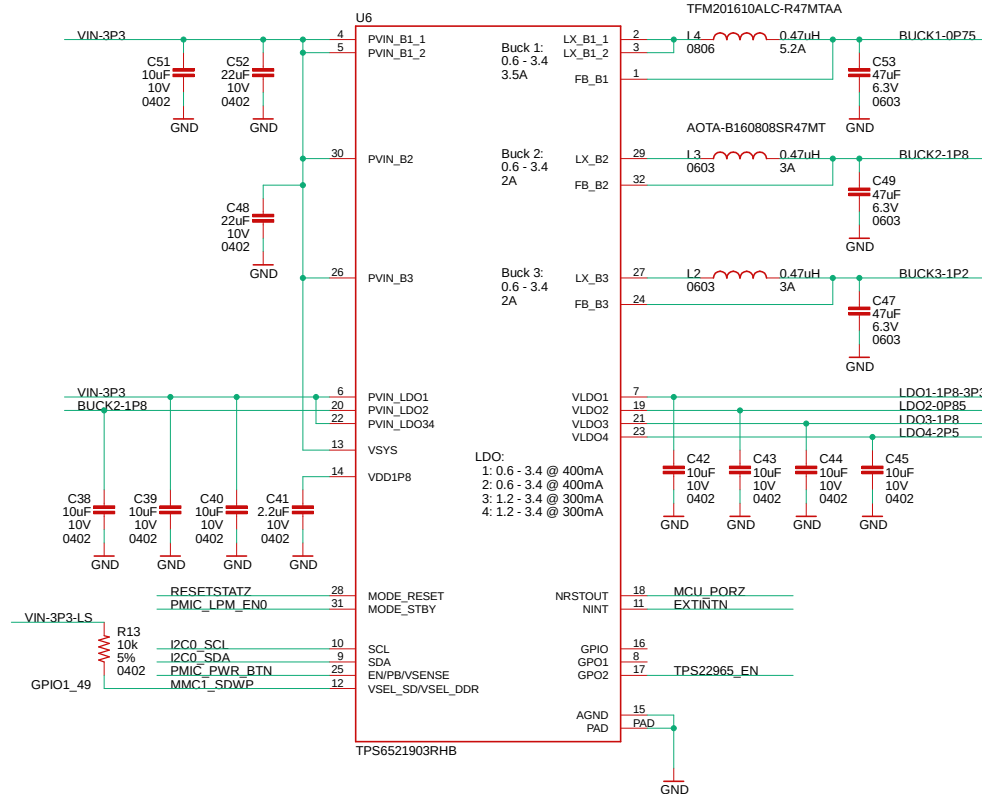
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# Power Management

See <https://www.ti.com/lit/an/slvafd0a/slvafd0a.pdf>  
- Using TPS6521903

I2C Address: 0x30 (7-bit)



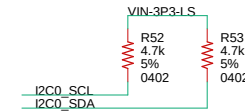
PMIC\_VSEL\_SD requires an external pull up to the voltage of the IO bank that drives it. This is VDDSHV0, i.e. VIN-3P3-LS, (MMC1\_SDWP) for this design.

GPO2 is enabled in the TPS6521903 configuration and used by the PMIC to enable the 3.3V load switch.

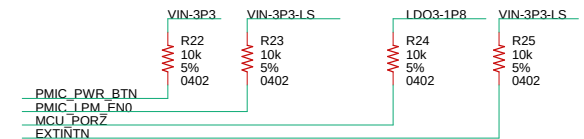
See <https://www.ti.com/lit/ug/slvucj2a/slvucj2a.pdf> for more information on the "03" configuration of the TPS65219.

See [https://octavosystems.com/app\\_notes/osd62x-pm-power-application-note/](https://octavosystems.com/app_notes/osd62x-pm-power-application-note/) for an in-depth discussion of the TPS65219.

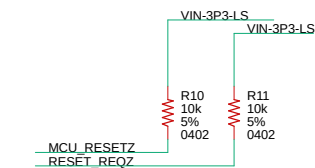
## I2C0 Pull-up Resistors



## Pull-up Resistors



## AM62 Reset Resistors



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## A

B

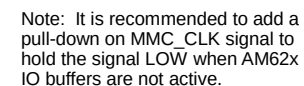
C



## E

## USB Connector

DFP Advertisement:  
Default USB Power (500mA/900mA)

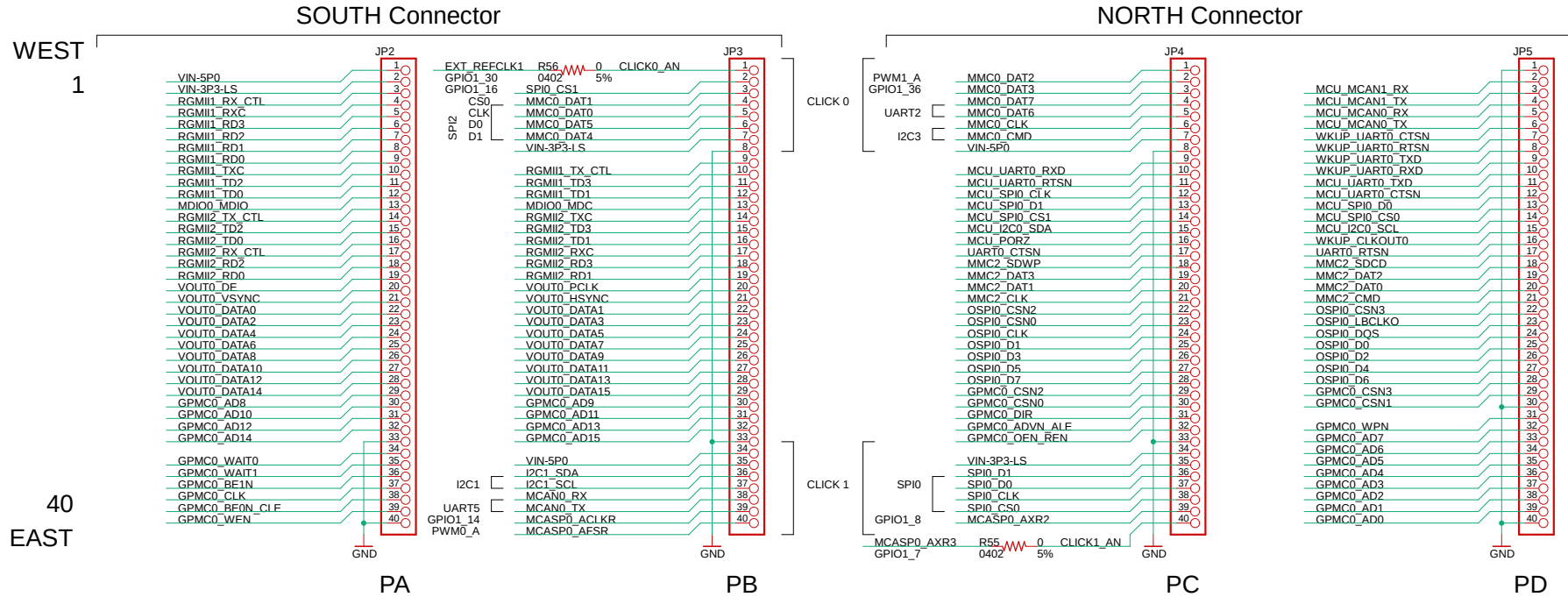


Rev: 1D



# IO Headers

Connectors support two (2) Mikroe Click Interfaces on the inner headers (East / West)



Click Modules come with I2C pull ups on-board.  
Footprints added on I2C1 and I2C3 for convenience (see other page).

GPMC[0:15] are used as boot mode pins and have default pull up / pull down behavior. See "Boot Mode" section for more information.

To meet the Mikroe Click standard, a resistor mux is used to add GPIO functionality to the CLICK0\_AN and CLICK1\_AN pins. This is generally not needed in a design based on the OSD62-PM-BRK.



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# OLDI / LVDS Display Interface

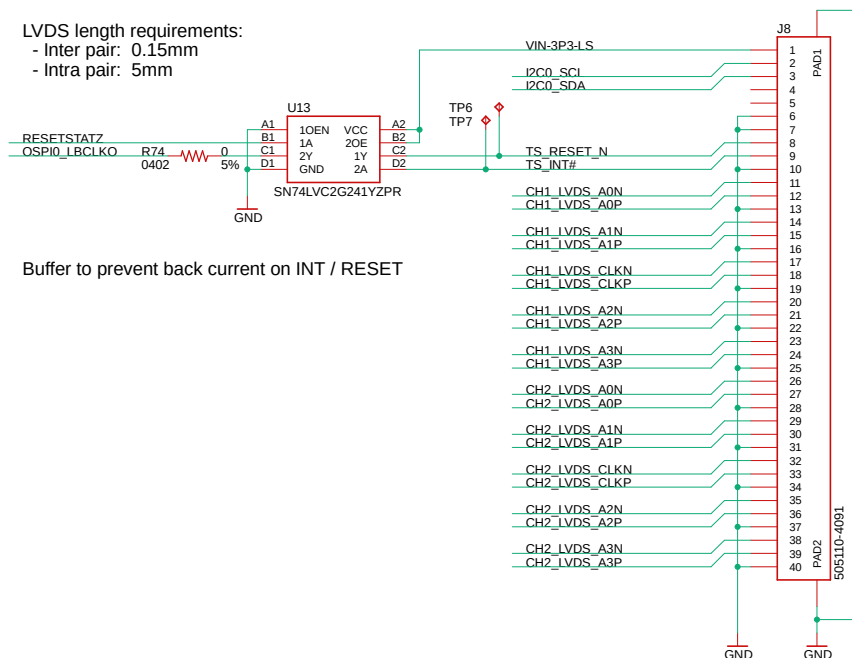
Based on AM62x SKEVM with Fulton PMIC

Supports: SK-LCD1 Interface (<https://www.ti.com/tool/SK-LCD1>)

Need a 40 pin FPC cable 0.5mm pitch (A Type) cable to connect to the screen.

LVDS length requirements:

- Inter pair: 0.15mm
- Intra pair: 5mm



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|   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|
|   | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
| A |   |   |   |   |   |   |   |   |
| B |   |   |   |   |   |   |   |   |
| C |   |   |   |   |   |   |   |   |
| D |   |   |   |   |   |   |   |   |
| E |   |   |   |   |   |   |   |   |
|   | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |

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# Design Notes

Revision 1A  
- Initial Revision - Pre-Production

Revision 1B  
- Fixed issue with USB connector shorting power and ground  
- Updated SD card connections to fix UHS-I speed transition  
- Updated PC / PD headers

Revision 1C  
- Added inverter on microSD card CD pin  
- Added back current protection for USB-C UFP connector  
- Added LVDS connector  
- Moved all components to top side of the board (except LVDS connector)

Revision 1D  
- Added buffer for UART  
- Added buffer for LVDS  
- Updated LVDS documentation  
- Added LDO for VPP per AM62x SKEVM



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